Name: Date:

Standards: MA.5.DP.1.1-1.2; MA.5.GR.4.1-4.2

Show your work in the box for each problem.

Write your final answer on the answer line (or in the blanks shown).

Use the words and strategies from Unit 12: data analysis and graphs.

1 Recall: Mean

Explain what the mean (average) of a data set represents.

Write 1 to 2 sentences.

The mean is the total of all values divided equally among all the data points — it represents the fair-share value for the data s

2 Recall: Coordinate Graph

In a coordinate graph, what is the difference between the x-axis and the y-axis?

Describe each axis.

The x-axis is the horizontal axis (left–right); the y-axis is the vertical axis (up–down). A point is written as (x, y).

3 Application: Mean

Six friends earned: \$12, \$15, \$9, \$18, \$14, \$10 at a bake sale.

Find the mean amount earned.

(12+15+9+18+14+10) ÷ 6 = 78 ÷ 6 = \$13.

4 Application: Line Plot

Hours of sleep per night for 8 students: 7, 8, 9, 8, 7, 9, 9, 8.

Create a line plot, identify the mode, and find the mean.

Mode = 8 and 9 (each appears 3 times). Mean = 65 ÷ 8 = 8.125 hours.

5 Word Problem: Savings

Mira saved $20, $35, $15, $40, and $30 over five months.

What was her average monthly savings? In which months did she save above average?

Mean = 140 ÷ 5 = $28. Above average: the $35 month and the $40 month.

6 Word Problem: Coordinate Graph

Plot A(3, 5), B(7, 5), C(7, 1), D(3, 1) on a coordinate graph.

What shape is formed? Find the area.

Rectangle. Width = 4 units; Height = 4 units. Area = 4 × 4 = 16 square units.

7 Word Problem: Straight Line Graph

A car rental costs $30 per day.

Write the cost for 1 to 5 days. What would 8 days cost?

1d=$30, 2d=$60, 3d=$90, 4d=$120, 5d=$150. 8 days = $30 × 8 = $240.

8 Find the Mistake

Leo calculated the mean of 5, 10, 15, 20 as: (5+10+15+20) ÷ 3 = 50 ÷ 3 ≈ 16.7.

Find Leo's error and calculate the correct mean.

Leo divided by 3 instead of 4. Correct: 50 ÷ 4 = 12.5.

9 Multi-Step: Raise the Average

Atlas scored 72, 80, 68, and 76 on four tests. He wants an average of at least 76 after five tests.

What is his current average? What minimum score does he need on the fifth test?

Current total = 296. Current average = 74. Target total = 76 × 5 = 380. Needed score = 380 – 296 = 84.

10 Reasoning: Mode vs. Mean

Shoe sizes: 7, 8, 8, 8, 9, 10, 12. Mode = 8; mean ≈ 8.86.

A shoe store owner wants to know which size to stock most. Should she use the mode or the mean? Explain.

She should use the mode (8), because it tells her the most commonly purchased size. The mean (8.86) does not correspond